

OPERATOR'S MANUAL

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CHAPTER FIVE

INSTALLATION AND ALIGNMENT

Introduction

In this chapter system installation requirements, and option alignment procedures, for the Infinity laser system are presented. The user should consult these sections if the laser system has or will be moved. When the system is delivered from Coherent, and set up for the first time, these procedures will be performed by the field service engineer. The user does, however, need to make some preparations before this installation. These are outlined below and in the Infinity Preinstallation Manual. Refer to Chapter Four, Daily Operation, for start-up and system troubleshooting procedures.

Setting up the Work Space

The Infinity laser system is a precision instrument whose performance depends on the properties of its environment. Vibrations can be transmitted to the laser head from the surface on which it rests, causing both beam pointing instability and power amplitude fluctuations. For the best results, the laser head should be placed on a stabilized optical table and isolated from mechanical contact with other equipment. Enough space should be left around the laser head so that the components inside can be reached easily for cleaning and maintenance.

The laser head and power supply are connected through a ten-foot umbilical cord. The laser head should be positioned so that the umbilical cord can hang freely. Allow at least 30 cm (12 inches) behind the power supply so that the umbilical cord will not be crimped and a minimum of 15 cm (6 inches) on the sides of the power supply, as well as the laser head, to ensure unobstructed air circulation. Figure 2-1 (Chapter Two, Description and Specifications) illustrates the recommended layout for the laser system.

Infinity Dimensions

Table 5-1. Dimensions of Infinity System

ITEM	LENGTH	WIDTH	HEIGHT
LASER HEAD	95 cm / 37.4"	48.6 cm / 19.2"	24.5 cm / 9.7"
POWER SUPPLY	68.6 cm / 27"	80 cm / 31.5"	53.4 cm / 21"
EMI FILTER	44 cm / 17.4"	36.5 cm / 14.4"	15.5 cm / 6.1"

Infinity Weights

The weight of the Infinity laser head is 150 lb (68 kg). The weight of the Infinity power supply without water is 235 lb (107 kg). With water the weight of the power supply increases to 272 lb (123 kg).

Ambient Temperature

The room in which your laser is located should be maintained at a temperature of 10-40°C (50-104°F). Stable operating temperature for the laser is around 30°C.

Cooling Water

The Infinity power supply incorporates a heat exchanger that regulates the temperature of a closed (“secondary”) water loop through the laser head. The heat exchanger requires a flow of “primary” cooling water from a tap or other water source. The cooling water should meet the specifications given in Table 5-2 and Figure 5-1.

A Flow rate valve, located inside the power supply, regulates the amount of water consumed. Primary water flow rate may vary from 3.8 liters (1 gallon) per minute if the water temperature is very cold, 10°C (50°F), to 22.7 liters (6 gallons) per minute, if the water temperature is warm (maximum 30°C (90°F)), see Figure 5-1. For best laser performance the water should not fluctuate more than $\pm 1^{\circ}\text{C}$.

Two 7.6 meter (25 foot) lengths of reinforced garden hose with female connectors are supplied with the laser. These are connected at the rear of the power supply and to facility inlet and drain. Remember to attach the bracket, that will hold the umbilical, to the power supply first. After the hoses have been connected, remove the larger of the two power supply covers and open the main facility valves, drain first, and inspect the system for leaks.

Table 5-2. Cooling Water Specifications

PARAMETER	DESCRIPTION	
Flow rate (min.):	20 lpm @ 30°C	5 gpm @ 86 F
Incoming Temperature:		
Minimum	5 ° C	41 ° F
Maximum	30 ° C	86 ° F

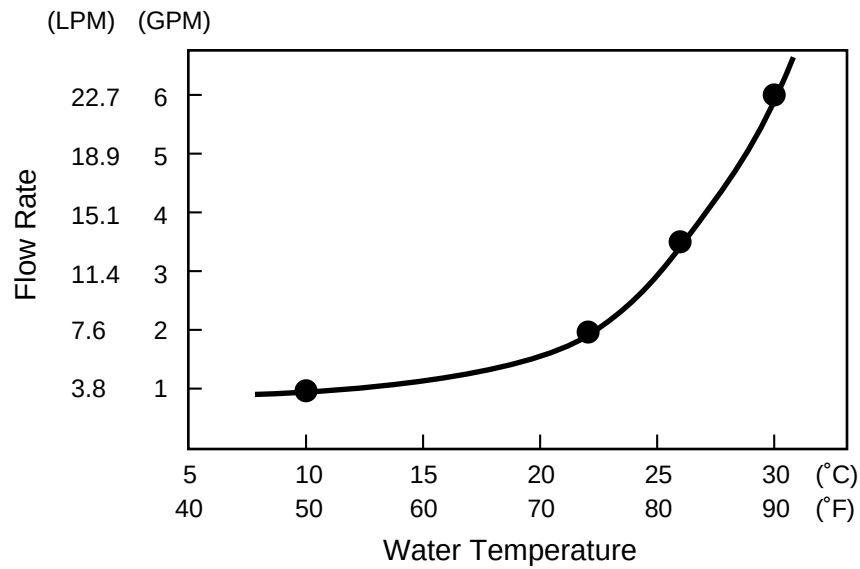


Figure 5-1. Primary Water Flow vs. Primary Water Temperature

External Heat Exchanger Requirements

An external closed loop heat exchanger used with the Infinity system needs to have a cooling capacity of at least 7.5 kW. This requirement is based on operation maximum repetition rate and at maximum lamp voltage, 550 V. Water temperature and flow specifications for the external cooling unit must be consistent with the requirements in Table 5-2.

Electrical Service

The AC electrical service for the Infinity power supply must meet the specifications in Table 5-3.

The electrical service may be either WYE or Delta connected, with the fourth wire to building ground. WYE is more tolerant to one line versus ground fluctuations and is therefore recommended. For safety reasons the circuit breaker should be located in the same room as the laser.

Attached to the power supply is a ten-foot power cable without a connector on the free end. The Infinity power cable is made up of 4x8G wires. The user must provide the hardware necessary to connect this cable to the local power service (e.g., Hubbell 4100P9W connector).

Table 5-3. Electrical Specifications

PARAMETER	DESCRIPTION	
Frequency (Hz)	60	50
Voltage (VAC),3-phase	208	380
Current-nominal (A)	20	15

System Computer

The Infinity laser is operated from an external (IBM compatible) computer using software from Coherent. Refer to Chapter Seven, System Software, for installation and hardware requirements.

The PC computer, on which the Infinity interface runs, connects to the RS 232 port on the power supply by a standard 9 pin cable. The COM port to which the cable is connected on the computer can be chosen in the Infinity software, see Chapter Seven.

Installation of External Interlock

The Infinity system provides the user with an external interlock circuit. It is located at the rear of the power supply, Figure 5-2. The system can only be turned on if this contact is short circuited. When the system is installed a short circuited RCA connector is provided for this purpose.

If it is desirable to connect the Infinity interlock to an external circuit an external switch which energizes a double pole single throw relay can be used, Figure 5-2.

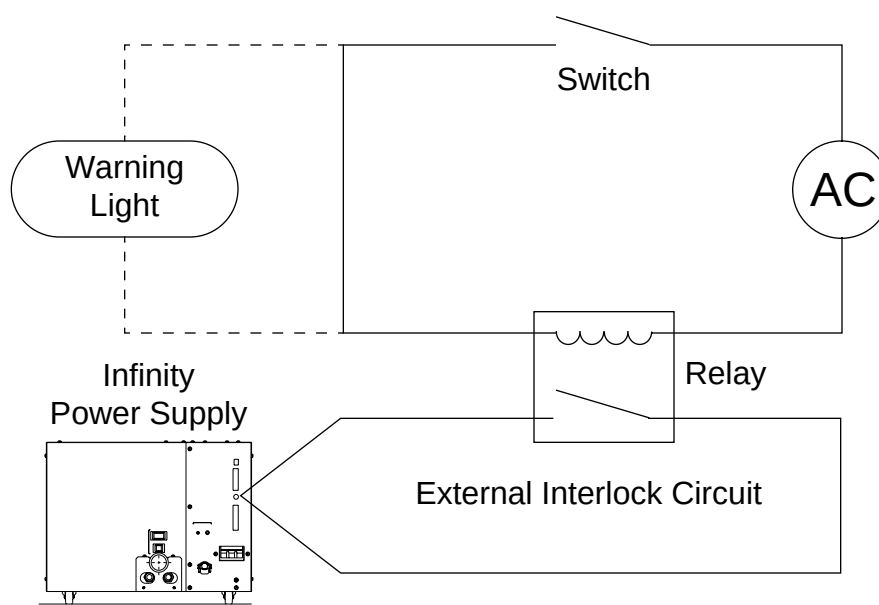
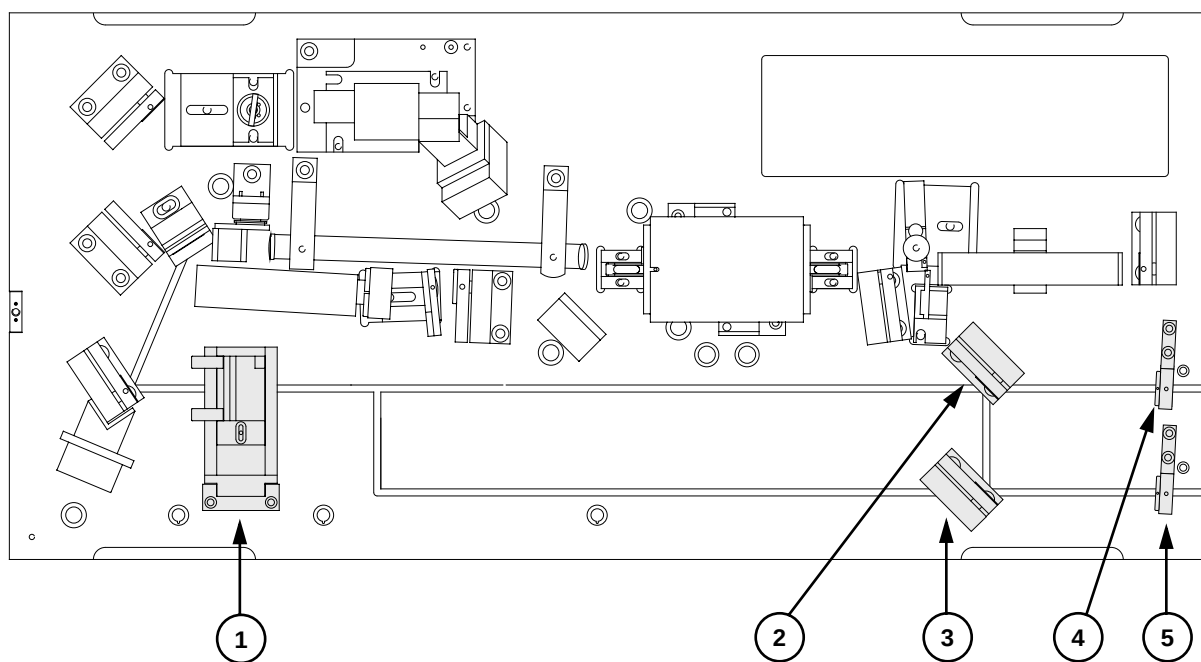


Figure 5-2. Connection of Infinity External Interlock to External On/Off Switch

Installation of SHG package

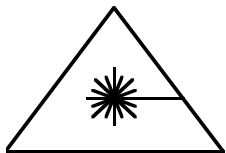
Initial installation of the Second Harmonic Generation package is done when the system is delivered or, if added later, by a Coherent field service engineer. If the package has been removed follow the procedure below to reinstall the SHG crystal. It is assumed that the necessary mounts and dichroics have been installed, see Figure 5-3.

If the residual IR is not to be utilized, the Infinity beam dump should be positioned in the location of the 1064 nm half waveplate, item 4 in Figure 5-3.



- 1 SHG assembly
- 2 Dichroic mirror.
Transmits 1064 nm and reflects 532 nm.
- 3 Dichroic mirror.
Transmits 1064 nm and reflects 532 nm.
- 4 Half wave plate for 1064 nm (optional)
- 5 Half wave plate for 532 nm (optional)

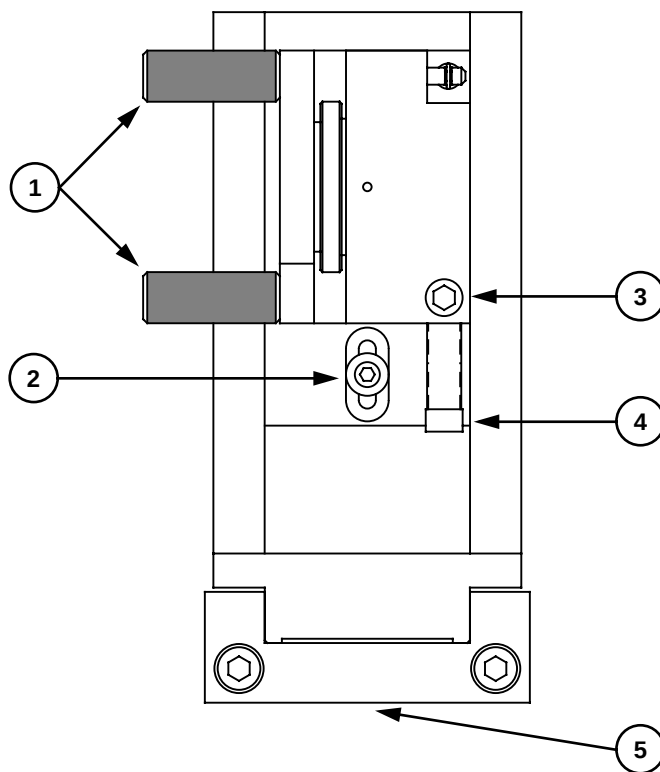
Figure 5-3. Layout for Second Harmonic Generation



The procedure described below requires the laser to be operated with the laser head cover open and the interlock defeated. Safety requires that everyone present in the room wears appropriate safety glasses and that access to the room be restricted.

SHG Installation Procedure

1. Open the head cover and install the interlock defeat. Place a power meter and expanding lens (e.g., $f = -10$ cm AR coated for 1064 and 532 nm) outside the Infinity head.
2. Run the Infinity control program. Set the repetition rate to 10 Hz, the IR pump energy to 1 mJ, and click on "On".
3. Using an IR viewing card, verify that the beam travels along the marked beam path on the base plate and at constant height above it (i.e., the pump beam is centered in the inboard exit aperture of the Infinity head cover). If not, use the vertical and horizontal tilt adjustments of the IR fold mirror assembly to position the pump beam.
4. Click on Standby. Install the alignment aperture in the cell holder. Restart the laser by clicking on On.
5. Adjust the horizontal and vertical tilt controls of the cell holder, 3 and 4 in Figure 5-4, so that aperture surface faces the incoming IR beam at normal incidence. Thereafter tilt the mount slightly around its horizontal axis so that a reflection from the cell window, when it is inserted, will strike the white diffuser, mounted above the IR fold mirror assembly. This is the position in which the crystal will be used and therefore the translational alignment should be carried out.
6. Increase the pump energy to 25 mJ. Adjust the vertical and horizontal translation stages on the SHG assembly, 2 and 5 in Figure 5-4, so that the beam is centered on the aperture.
7. Increase the power in the main control window in steps of 25 mJ. If scattered light appears anywhere along the edges of the aperture move the cell holder until the beam clears the aperture. Continue until 600 mJ can be transmitted. Use burn paper to verify that the beam is not being clipped by the aperture.



- 1 Lock nuts securing the cell to the gimbal mount
- 2 Lock screw for horizontal translation
- 3 Vertical adjustment screw controlling tilt around horizontal axis
- 4 Horizontal adjustment screw controlling tilt around vertical axis
- 5 Lock screw for vertical translation

Figure 5-4. Crystal Holding Assembly

8. Set the IR energy to 1 mJ and click on Standby.
9. Remove the alignment aperture and install the SHG crystal. Position a white paper in the 532 nm beam path.
10. Click on On and set the IR power to 25 mJ.
11. Tilt the crystal back and fourth, by hand, around its **vertical** axis. Look for a bright green spot to appear on the white paper. Use the horizontal control screw, controlling the tilt around the vertical axis (4 in Figure 5-4), to set the angle at which the 532 nm light appears with maximum brightness.

12. After the optimal phase-matching angle has been found, if required, tilt the crystal around its horizontal axis (3 in Figure 5-4) so that the return reflex from the entrance window strikes the white diffuser above the IR fold mirror assembly.



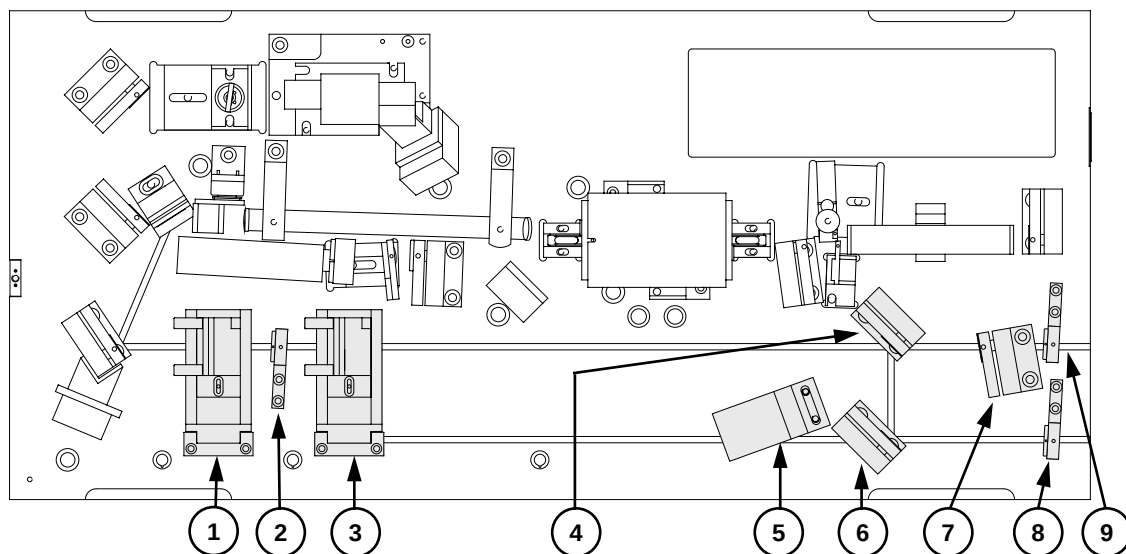
If the beam travels through the crystal off center irreversible damage may occur. Make sure at low powers that the 532 nm beam goes through the center of the crystal and that it is round and symmetrical.

13. Set the IR energy to 1 mJ and click on Standby. Remove the SHG crystal and reinstall the alignment aperture. Verify that the pump beam is still centered in the aperture. If it is not, the necessary corrections should be made (see above). This is done to ensure that the crystal is still centered on the beam after it has been tilted.
14. At low power, ~25 mJ, adjust the tilt angles of the first dichroic so that the green beam strikes the second dichroic in the center. Adjust the controls of the second dichroic so that the beam exits through the center of the outboard exit aperture in the Infinity head cover.
15. With the laser running at the desired repetition rate and pump energy, optimize the SHG phase matching angle.
16. Turn off the system, remove the interlock defeat and close the head cover. The system is now ready to be used.

Installation of THG package

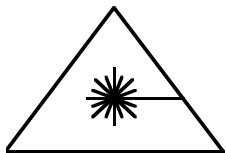
Initial installation of the Third Harmonic Generation package is done when the system is delivered or, if added later, by a Coherent field service engineer. If the package has been removed follow the procedure below to reinstall the THG crystal. It is assumed that the SHG option and the necessary mounts and dichroics have been installed, see Figure 5-5.

If the residual 532 nm radiation is not to be utilized, the Infinity beam dump (item 5 in Figure 5-5) should be positioned in the location of the dichroic fold mount and 532 nm half wave plate (items 7 and 9 in Figure 5-5).



- 1 SHG assembly
- 2 Wave plate
- 3 THG assembly
- 4 Dichroic mirror.
Transmits 1064 nm and 532 nm and, reflects 355 nm.
- 5 Beam dump for 1064 nm
- 6 Dichroic mirror.
Transmits 1064 nm and 532 nm and, reflects 355 nm.
- 7 Dichroic mirror (optional).
Transmits 532 nm and reflects 1064 nm.
- 8 Half wave plate for 355 nm (optional)
- 9 Half wave plate for 532 nm (optional)

Figure 5-5. Layout for Third Harmonic Generation



The procedure described below requires the laser to be operated with the laser head cover open and the interlock defeated. Safety requires that everyone present in the room wears appropriate safety glasses and that access to the room be restricted.

Installation Procedure for THG

1. Make sure that the two dichroics, 4 and 6 in Figure 5-5, are for 355 nm. Place a power meter and expanding lens (e.g., $f = -10$ cm AR coated for 532 and 355 nm) outside the Infinity head.
2. Run the Infinity control program. Set the repetition rate to 10 Hz, the IR pump energy to 1 mJ, and click on “On”.
3. Position the waveplate holder, 2 in Figure 5-5, relative to the beam according to Figure 5-6 and Figure 5-7. Secure it to the base plate.

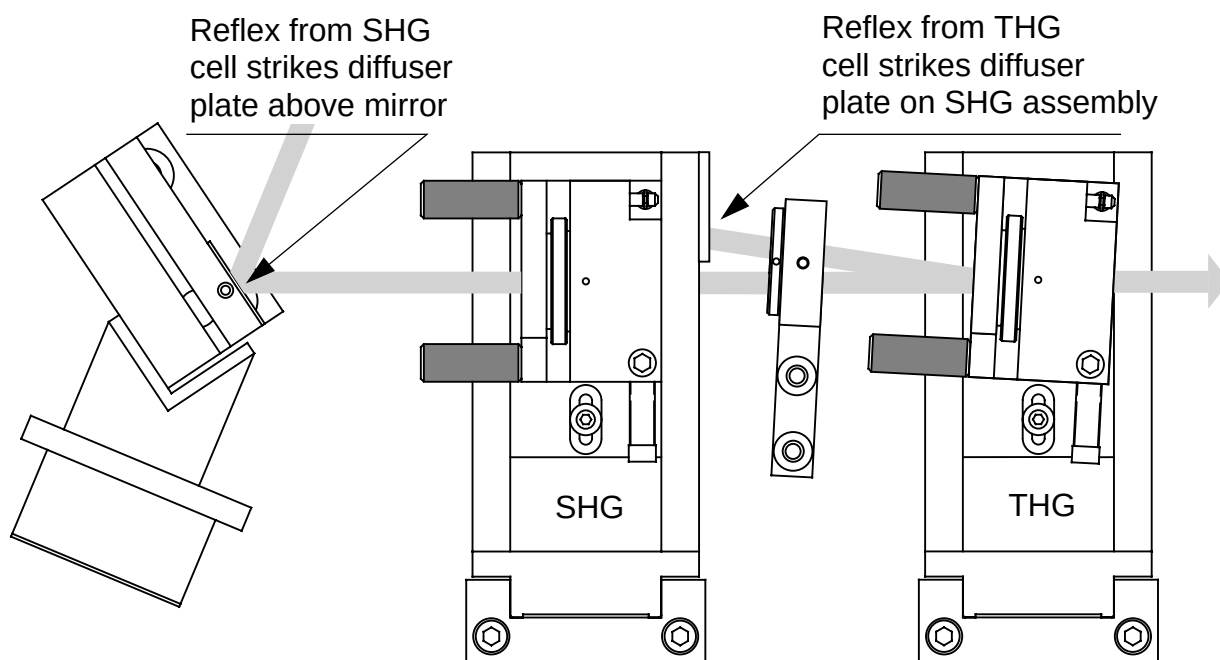


Figure 5-6. Beam Path through SHG and THG Optics

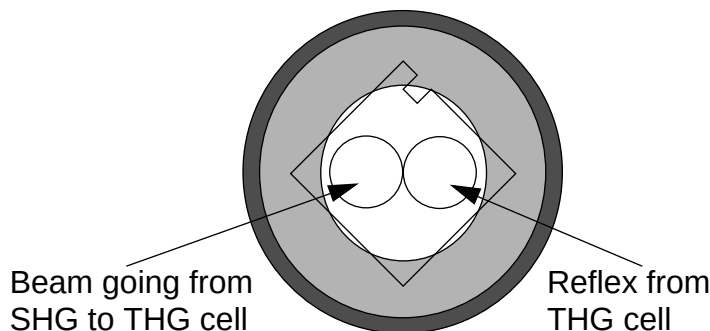


Figure 5-7. Beam Paths through THG Wave Plate, seen from the THG Crystal

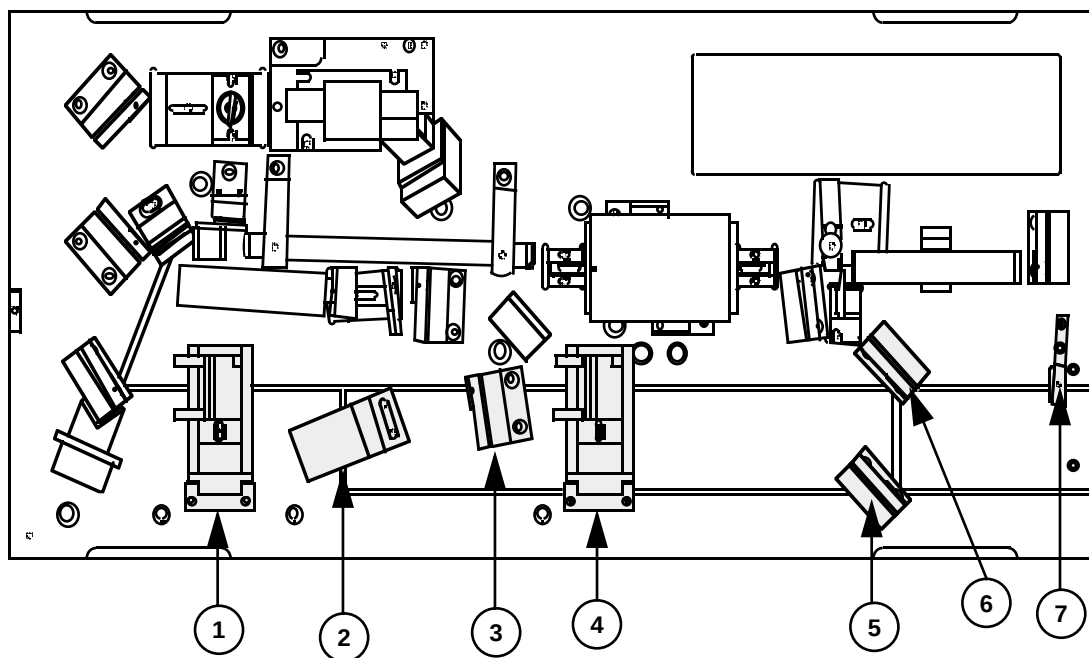
4. Click on Standby. Install the alignment aperture in the crystal cell holder. Restart the laser by clicking on On.
5. Adjust the horizontal and vertical tilt controls of the cell holder, 3 and 4 in Figure 5-4, so that aperture surface faces the incoming 1064/532 nm beam at normal incidence. Thereafter tilt the cell slightly around its vertical axis so that the reflection from the cell window, when the crystal is inserted, will strike the white diffuser, mounted on the side of the SHG assembly. This is the cell position in which the crystal will be used and therefore the alignment should be carried out.
6. Adjust the vertical and horizontal translation stages on the THG assembly, 2 and 5 in Figure 5-4, so that the beam is centered on the aperture.
7. Increase the power in the main control window in steps of 25 mJ. If scattered light appears anywhere along the edges of the aperture move the cell holder until the beam clears the aperture. Continue until 600 mJ can be transmitted. Use burn paper to verify that the beam is not being clipped by the aperture. Compare burn patterns taken before and after the aperture.
8. Turn down the power to 1 mJ and click on Standby.
9. Position a white paper in the 355 nm beam path.
10. Remove the alignment aperture and install the THG crystal.
11. Restart the laser by clicking on On. Set the IR power to 25 mJ. Optimize the intensity of the green light by adjusting the SHG crystal angle. Increase the IR power to 50 mJ.

12. Tilt the THG crystal back and fourth, by hand, across its full travel range around the **horizontal** axis. Look for a bright blue fluorescent spot to appear on the paper. Use the vertical control screw, controlling the tilt around the horizontal axis, to set the angle at which the blue light appears with maximum brightness.
13. After the phase-matching angle has been found, tilt the crystal around its vertical axis so that the return reflex from the entrance window strikes the white diffuser.
14. Set the IR energy to 1 mJ and click on Standby. Remove the THG crystal and install the alignment aperture. restart the laser by clicking on On. Repeat steps 6–9. This is done to ensure that the crystal is still centered on the beam after it has been tilted.
15. Adjust the tilt angles of the first dichroic so that the tripled beam strikes the second dichroic in the center. Adjust the tilt angles of the second dichroic so that the tripled beam exits through the center of the outboard exit aperture of the Infinity head cover.
16. Gradually increase the power while monitoring the (green) mode on the crystal surface. Adjust the phase matching angle so that the blue light appears with maximum brightness. If it becomes obvious, as the laser power is turned up, that the beam does not go through the center of the crystal go back and repeat the alignment procedure from step 2.
17. Make small rotational (a few degrees) adjustments to the wave plate until the energy at 355 nm is maximized.
19. In the tripling process one photon at 1064 nm is mixed with one at 532 nm. Optimum efficiency for the tripling process is therefore achieved when there is one photon at 532 nm available for every photon at 1064 nm, a 50/50 mix. To accomplish this it might be necessary to slightly detune the SHG crystal from the position at which the SHG efficiency is the highest. With the laser running at full power, optimize the tripled energy by slightly detuning the SHG crystal. Finally optimize the THG phase matching angle.
20. Turn off the system, remove the interlock defeat and close the head cover. The system is now ready to be used.

Installation of FHG package

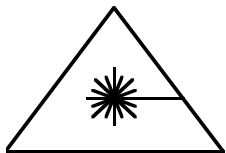
Initial installation of the Fourth Harmonic Generation package is done when the system is delivered or, if added later, by a Coherent field service engineer. If the package has been removed follow the procedure below to reinstall the FHG crystal. It is assumed that the SHG option and the necessary mounts and dichroics have been installed, see Figure 5-8.

If the residual 532 nm radiation is not to be utilized, the second Infinity beam dump should be positioned in the location of the 532 nm half wave plate (item 7 Figure 5-8).



- 1 SHG assembly
- 2 Beam dump for 1064 nm
- 3 Strip out dichroic
- 4 FHG assembly
- 5 HR 266 nm dichroic mirror
- 6 Dichroic mirror
Transmits 532 nm and reflects 266 nm
- 7 Half wave plate for 532 nm (optional)

Figure 5-8. Layout for Fourth Harmonic Generation



The procedure described below requires the laser to be operated with the laser head cover open and the interlock defeated. Safety requires that everyone present in the room wears appropriate safety glasses and that access to the room be restricted.

Installation Procedure for FHG

1. Open the Infinity head cover and install the interlock defeat. Install the stripout dichroic mount (item 3 in Figure 5-8) without the dichroic, the FHG crystal mount (item 4 in Figure 5-8) and the second beam dump (item 2 in Figure 5-8). Install the alignment aperture in the FHG crystal mount.
2. Place a power meter outside the Infinity head, in-line with the outboard beam path on the Infinity baseplate. Run the Infinity control program. Set the repetition rate to 10 Hz, the IR pump energy to 1 mJ, and click on "On".
3. Adjust the horizontal and vertical tilt controls of the FHG cell holder, items 3 and 4 in Figure 5-4, so that aperture surface faces the incoming IR beam at normal incidence. Thereafter tilt the cell slightly around its vertical axis so that the reflection from the cell window, when the crystal is inserted, will strike the white diffuser, mounted on the side of the SHG assembly, see Figure 5-9. This is the cell position in which the crystal will be used and therefore the alignment should be carried out.



The back reflection from the FHG cell needs to be aligned so that it will go through the IR stripout dichroic (when installed) without clipping the dichroic mount.

4. Adjust the vertical and horizontal translation stages on the FHG assembly, items 2 and 5 in Figure 5-4, so that the beam is centered on the aperture.

5. Increase the power in the main control window in steps of 25 mJ. If scattered light appears anywhere along the edges of the aperture move the cell holder until the beam clears the aperture. Continue until 600 mJ IR power has been reached. Use burn paper to verify that the beam is not being clipped by the aperture. Compare burn patterns taken before and after the aperture.
6. Turn down the Infinity IR energy to 1 mJ and click on Standby. Install the IR stripout dichroic and the FHG crystal, mounts 3 and 4 in Figure 5-8.
7. Set the repetition rate to 10 Hz and click on On. Increase the IR energy in 1 mJ steps until a distinct IR beam can be seen on the IR beam dump, item 2 in Figure 5-8. Reposition the beam dump and /or the strip out dichroic mount if necessary.

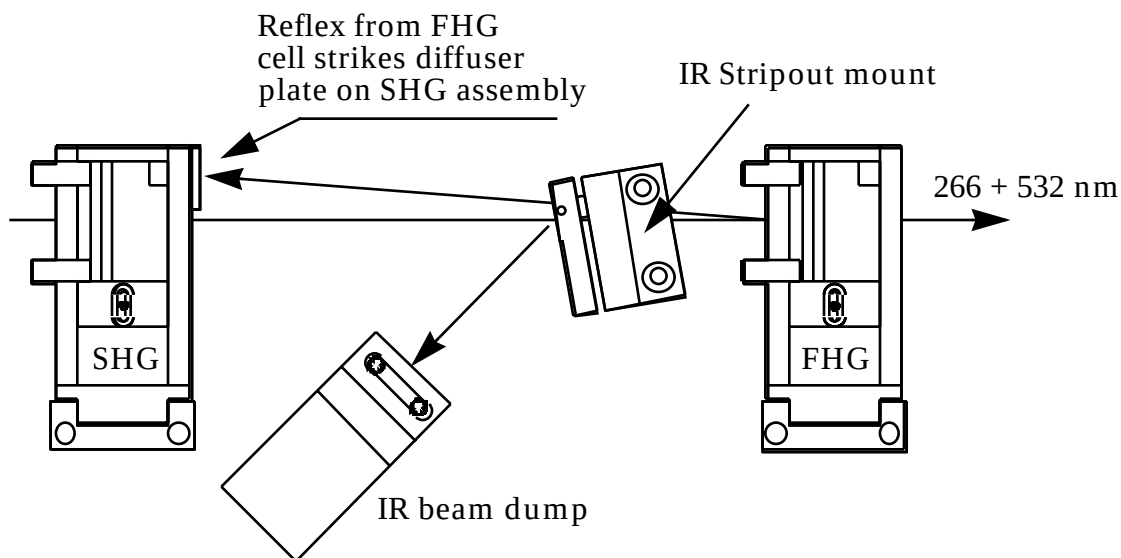


Figure 5-9. Beam Path through SHG and FHG Optics

8. Look for the 532 nm back reflection on the ceramic diffuser attached to the SHG mount. Adjust the vertical tilt of the FHG mount to position the back reflection on the diffuser if necessary.

9. Verify that both the SHG output beam, and that the back reflection from the FHG crystal, are not being clipped by the IR stripout dichroic mount.
 - a. If SHG beam is being clipped, the IR stripout mount will need to be repositioned.
 - b. If the back reflection from the FHG cell is being clipped the tilt angle of the harmonic mount will need to be corrected.

If large adjustments to the FHG mount are required, reinstall the alignment aperture and return to step 4.



The standard beam expanders can not be used with the FHG option. It will therefore be necessary to send the 266 nm output directly into a power meter head. This head should be equipped with a UV diffuser to prevent damage to the detector.

10. Place the system in Standby. Place a white card, or paper, in the beam path. Restart the laser by clicking on ON and increase the IR power to 50 mJ.
11. Tilt the FHG crystal back and forth, by hand, across its full travel range around the **horizontal** axis. Look for a blue fluorescent spot to appear on the white paper. Use the vertical control screw, controlling the tilt around the horizontal axis (item 3 in Figure 5-4), to set the angle at which the 266 nm spot appears with maximum brightness.
12. After the phase-matching angle has been found, again verify that the return reflex from the FHG entrance window strikes the white diffuser mounted on the side of the SHG crystal assembly.
13. Verify that the back reflection is not being clipped by the IR stripout dichroic mount.
14. Set the IR energy to 1 mJ and click on Standby. Remove the FHG crystal and install the alignment aperture. Restart the laser by clicking on On. Repeat steps 4 and 5 with the stripout dichroic installed. This is done to ensure that the crystal is still centered on the beam after it has been tilted.

15. Adjust the tilt angles of the first turning dichroic (item 6 in Figure 5-8) so that the 266 nm beam strikes the second dichroic in the center. Adjust the tilt angles of the second dichroic so that the 266 nm beam exits through the center of the outboard exit aperture in the Infinity head cover.
16. Gradually increase the power while monitoring the (green) mode on the crystal surface. Adjust the phase matching angle so that the blue light appears with maximum brightness. If it becomes obvious, as the laser power is turned up, that the beam does not go through the center of the FHG crystal, go back and repeat the alignment procedure from step 2.
17. To generate the Fourth Harmonic, two photons of 532 nm are required. Therefore, it really should not be necessary to go back and make any adjustments to the phase matching of the SHG crystal. However, the fact that the SHG crystal is optimized should be verified.
18. Turn off the system, remove the interlock defeat and close the head cover. The system is now ready to be used.

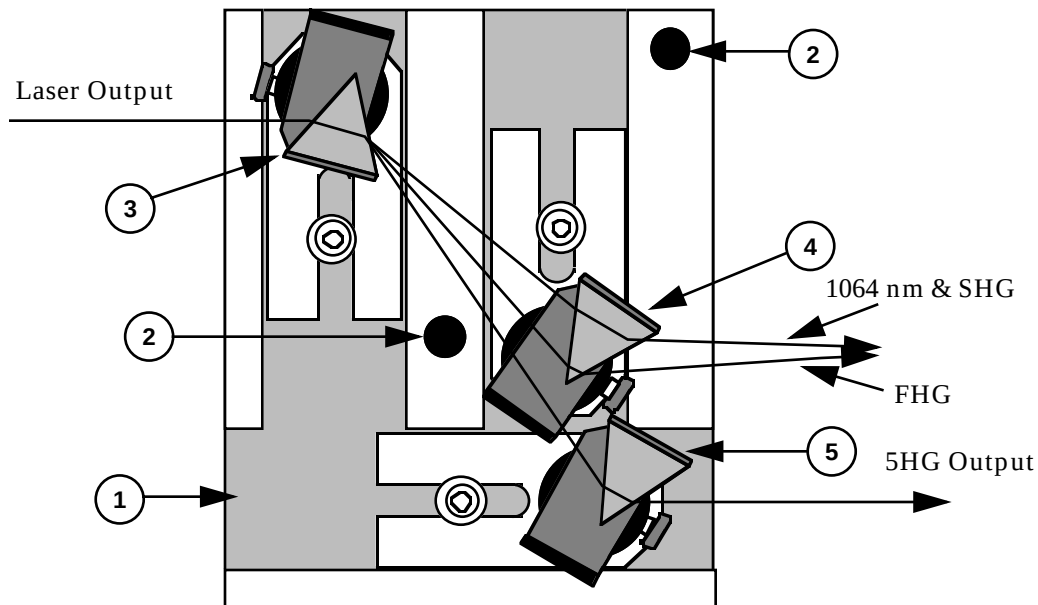


Due to the large nonlinear coefficient of BBO at 266 nm, 133 nm radiation can also be generated by the same crystal employed to generate the 266 nm radiation. BBO is not transparent to 133 nm radiation resulting in a significant temperature gradient as the pump energy and repetition rate of the laser system are changed. It is therefore necessary, and normal, to have to adjust the phase matching angle of the FHG crystal if the laser system operational parameters are changed by more than a few percent.

Installation of 5HG package

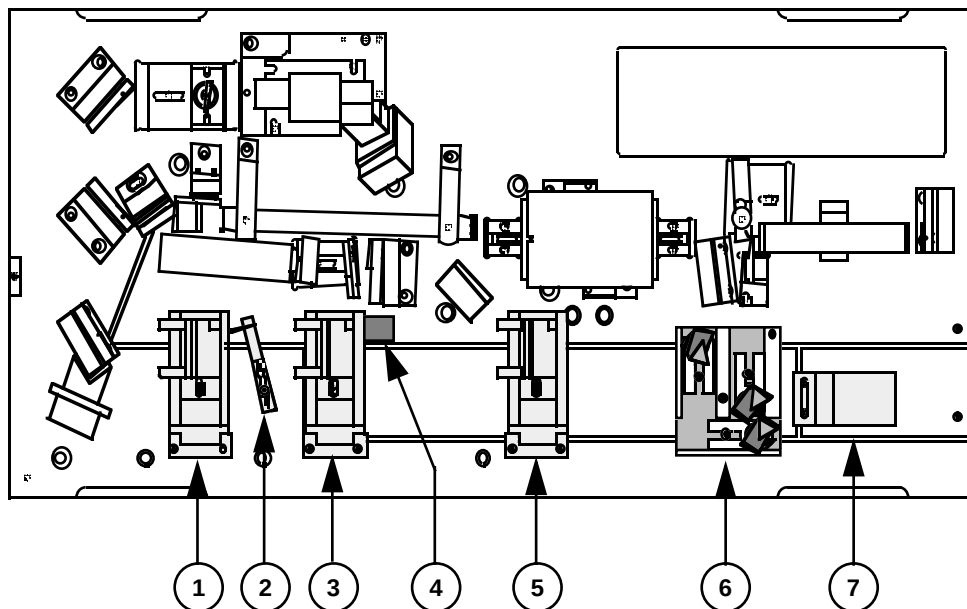
Initial installation of the 5TH Harmonic Generation package is done when the system is delivered or, if added later, by a Coherent field service engineer. If the package has been removed follow the procedure below to reinstall the 5HG crystal.

In the procedure which follows it is assumed that the system is operating in the FHG configuration. If not, install the FHG option first. Figure 5-10 gives an overview of the prism assembly employed to separate the 5TH harmonic from the 1064, 532 and 266 nm radiation. Figure 5-11 illustrates the proper mounting of the prism assembly as well as the 5TH harmonic crystal holder, the SHG and FHG retro-reflection ceramics, and the standard Infinity beam dump.



1	Base Plate
2	Securing Bolt Holes
3	Prism 1 (P1)
4	Prism 2 (P2)
5	Prism 3 (P3)

Figure 5-10. Prism Separation Assembly.



1	SHG Assembly
2	SHG Retro-reflection Ceramic
3	FHG Assembly
4	FHG Retro-reflection Ceramic
5	5HG Assembly
6	Prism Separation Assembly
7	Beam Dump

Figure 5-11. Layout for Fifth Harmonic Generation

Assembly Identification

Figure 5-12 gives a side and top view of a prism mount assembly. It is important to note the differences between the mounts, and to install each mount in its proper location. Depending on the version of your 5HG prism separation assembly there may be one or two differences between the prism mounts.

Common to all versions of this assembly is the fact that the mount for P1 is the mirror image of the mounts for P2 and P3 (see Figure 5-10). The second difference, found only in early versions of the assembly, is that the mount for P2 is longer (1-3/16") than the mount used for P3 (1"). This dimension is labeled "A" in the right hand illustration of Figure 5-12. In later versions of the assembly the mounts for P2 and P3 are the same length. The left hand illustration of Figure 5-12 indicates that a ceramic beam block is attached to each the

prism mounts. In Figure 5-10 these beam blocks were intentionally omitted for clarity in the drawing.

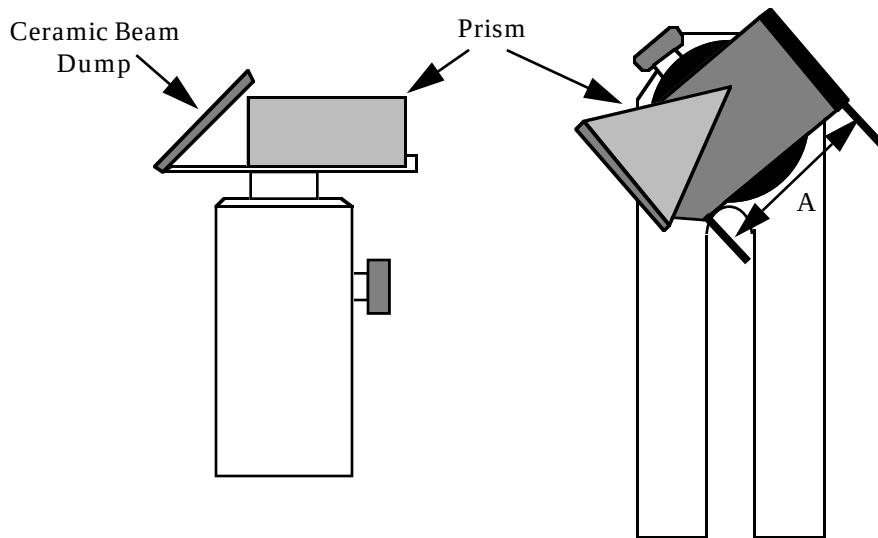


Figure 5-12. Prism Mount Assembly.

Installation Procedure for 5HG

To determine the angle between a prism face and the input beam, a card should be used to determine the position of the prism which results in the minimum deviation of the beam of interest (i.e., the position of the prism where the beam is least distorted along the horizontal axis). In this case, the beam of interest is the 213 nm beam. When the condition of minimum deviation is obtained the entrance and exit angles of the laser beam with respect to the prism faces will be identical and the largest possible aperture through the prism is realized. In this orientation the lowest possible reflection losses due to the prism faces is obtained.

1. Open the Infinity head cover and install the interlock defeat. Install the 5HG crystal cell holder and the alignment aperture.
2. If the 5HG option has not been previously installed, install the ceramic diffuser into its slide mount and attach the slide mount to the front of the FHG mount (see Figure 5-11). Note, if the FHG mount is an old style mount there will "not" be a screw hole for attachment of the ceramic slide mount. If this is the case the FHG harmonic mount will need to be switched with the 5HG mount. The ceramic should be positioned slightly to the

right of center (towards the amplifier stage) with respect to the center of the FHG crystal.

3. If the standard ceramic diffuser is attached to the SHG crystal stage (i.e., the diffuser employed for THG), remove this diffuser and install the FHG retro-reflection diffuser, item 2 in Figure 5-11.
4. Place a power meter outside the Infinity head, in line with the outboard beam path on the Infinity baseplate. Run the Infinity control program. Set the repetition rate to 10 Hz, the IR pump energy to 1 mJ, and click on “On”.
5. Position a beam dump, on the Infinity head baseplate, to collect the output beam (1064, 532 and 266 nm radiation). Increase the voltage until the threshold for FHG is reached. Position the FHG retro-reflection ceramic so that the beam is centered on the ceramic and is elongated to cover as large an area on the ceramic as possible, see Figure 5-13.

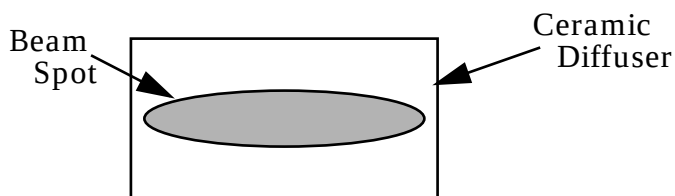


Figure 5-13. Beam Image on Properly Aligned Retro-Reflection Ceramic.

6. Adjust the horizontal and vertical tilt controls of the 5HG cell holder, 3 and 4 in Figure 5-4, so that aperture surface faces the incoming IR beam at normal incidence. Thereafter tilt the cell slightly around its horizontal axis so that the reflection from the cell window, when the crystal is inserted, will strike the ceramic diffuser, mounted on the FHG assembly. This is the cell position in which the crystal will be used and therefore the alignment should be carried out.
7. Adjust the vertical and horizontal translation stages on the 5HG assembly, 2 and 5 in Figure 5-4, so that the beam is centered on the aperture.
8. Increase the power in the main control window in steps of 25 mJ. If scattered light appears anywhere along the edges of the aperture move the cell holder until the beam clears the aperture. Continue until 600 mJ can be transmitted. Use burn paper to verify that the beam is not being clipped by the aperture. Compare burn patterns taken before and after the aperture.

9. Turn down the power and click on Standby.
10. Position P1 as illustrated in Figure 5-10. Place a white card, mounted on a beam dump, in the expected beam path due to P1 (i.e., near the location of P3 in Figure 5-10 or 5-11).
11. Remove the alignment aperture and install the 5HG crystal.
12. Restart the laser by clicking on “On” and increase the IR energy to 25 mJ. Position P1 so that the laser beam is striking the center of prism with respect to both the vertical and horizontal axis. Slowly increase the IR power until the 266 nm (FHG) radiation can be seen on the beam dump card as well as the 532 nm radiation. Increase the IR power until the 266 nm beam is stable (approximately 100 mJ IR energy). Optimize the phase matching angle of the FHG cell. Adjust the angle of P1 to minimize the deviation of the 532 and 266 nm beams.
13. Tilt the 5HG crystal back and fourth, by hand, across its full travel range around the **vertical** axis. Look for a second blue fluorescent spot to appear on the white card. Use the horizontal control screw, controlling the tilt around the vertical axis (4 in Figure 5-4), to set the angle at which the second blue spot appears with maximum brightness.
14. After the phase-matching angle has been found, tilt the crystal around its horizontal axis so that the return reflex from the entrance window strikes the white diffuser mounted below the FHG crystal assembly. If this is the first time the 5HG option is being installed, it may be necessary to reposition the ceramic. Re-establish the 5HG phase matching angle.
15. Set the IR energy to 1 mJ and click on Standby. Remove the 5HG crystal and install the alignment aperture. restart the laser by clicking on On. Repeat steps 7-14 to ensure that the crystal is still centered on the beam after the final tilting of the harmonic stage.

16. If necessary increase the IR energy to stabilize the 5HG beam. Adjust the angle of P1, with respect to the laser beam, to minimize the deviation of the “5HG” beam. Position P2 as illustrated in Figure 5-10. When properly installed the SHG beam should be at the top of the prism and the FHG near the bottom. The 5HG beam should clear the prism, and the prism mount, as illustrated in Figure 5-10. It may be necessary to raise the mount of P2 to get the beams centered vertically.
17. Rotate P2 to minimize the deviation of the 532 and 266 nm beams. Position the Infinity beam dump as illustrated in Figure 5-11. Verify that the 213 nm beam is not being clipped by P2 or its mount.
18. Position P3 as illustrated in Figure 5-10. Position the prism so that the deviation of the 213 nm beam is minimized and the beam exits cleanly out of the Infinity head cover aperture.



It is important to note that the normal 355 expansion lens can not be used for the 213 nm radiation generated by this option. It is therefore very important to use a (low energy) pulsed power meter equipped with a UV diffuser.

19. Verify that the power meter, equipped with a UV diffuser, in the path of the 213 nm beam. Gradually increase the power while adjusting the phase matching angle of the 5HG stage.



Typical energies for the 5TH harmonic are: 5 mJ at 100 Hz and 400 mJ IR and 15 mJ at 10 Hz and 600 mJ IR. The documentation which came with the 5HG option, or the laser system if the 5HG option was part of the original purchase, should be referenced to determine the energies specified by Coherent at the time of manufacturing.



Due to thermal loading, changes in the repetition rate or IR power by more than 5 to 10% may require that the phase matching angles of the FHG and 5HG crystals be optimized as well as the three prism angles (i.e., the minimum deviation of the SHG, FHG and 5HG beams). The latter can be most easily accomplished using a white card to follow the beam paths from one prism to the next, starting with P1.

20. To generate the 5TH harmonic, one photon of 266 nm is mixed with one at 1064 nm. Optimum efficiency for the process is therefore achieved when there is one photon at 266 nm available for every photon at 1064 nm, a 50/50 mix. To accomplish this it will be necessary to slightly retune the SHG and the FHG crystals from the position at which the FHG efficiency was the highest.

Set the repetition rate to the desired value (i.e., typical rate at which the system will be used). Set the IR energy to its maximum value. Optimize the 213 nm energy at these system parameters by slightly retuning the SHG crystal and then the FHG crystal. Finally, optimize the 5th harmonic phase matching angle.

Close the head cover and allow the system to run for approximately 10 minutes. Repeat the phase matching procedure until the maximum 213 nm energy is obtained. If less 213 nm radiation is desired decrease the IR energy before repeating the optimization procedure.

Installation of Telescope Assembly

The following is the procedure for installing and aligning the Infinity telescope. There are three different optic combinations which can be installed in the Infinity telescope. The optic numbers, focal lengths, AR coatings, and resulting magnification are given in Table 5-4. Note that optic “A” (installed in mount #1) is the optic closest to the harmonic stage and optic “B” (installed in mount #2) is the optic closest to the exit window in the Infinity head cover.

Table 5-4. Infinity Telescope Optics

	Optic “A”	Optic “B”	Coatings	Magnification
1	0167-912-00 (f = 148 mm)	0169-830-00 (f = 289 mm)	532 / 355	2x (11 mm)
2	0167-912-00 (f = 148 mm)	0167-913-00 (f = 270 mm)	532 / 355	1.8x (9.9 mm)
3	0167-910-00 (f = 148 mm)	0167-911-00 (f = 270 mm)	1064 / 532	1.8x (9.9 mm)

1. An overview of the installed telescope baseplate, location in head, is given in Figure 5-14. Align the mount baseplate so that the two attaching holes, indicated in Figure 5-15, line up with threaded holes in the Infinity baseplate. Secure the telescope mount baseplate to the Infinity head baseplate.
2. There will be a focused back reflection from the first positive lens of the Infinity telescope. This reflection will strike the second turning dichroic mount just below the optic. To prevent damage to both the mount and the dichroic two ceramic beam blocks should be stacked and mounted (double sided tape works well) just below the turning dichroic.
3. Construct two laser beam targets. Turn on the Infinity and adjust the IR energy until there is stable harmonic production. If an IR telescope is being installed then the IR power should be adjusted to 5 - 10 mJ.

Position one of the targets as far from the Infinity head as possible (4 to 5 meters). If necessary use external turning optics. Position the second target just outside the Infinity head. Both targets should be positioned so that the laser beam strikes the center of the targets.

4. The optics of the Infinity telescope are plano-convex. There is a penciled arrow on the side of each optic which points towards the convex side. The optics are mounted in the telescope so that the arrows (convex sides) point outward (away from the center of the telescope assembly).



The lens retaining ridge of lens mount #1 is vertically tilted. The telescope optic should be installed so that the convex side of the optic is orientated downwards and is facing towards the amplifier stage of the Infinity.

Install optic "A" into lens mount #1. Tighten the set screw to secure the lens. Adjust the vertical and horizontal position of the lens so that the beam is centered on the optic, and the beam position on the target closest to the Infinity laser head is not perturbed (i.e., the expanding beam is still striking the center of the target).

5. Install the vacuum cell on the telescope mount baseplate so that the window on the vacuum cell is as close to lens "A" as possible. This will prevent damage to the cell window from internal reflections in lens "A".
6. Install lens "B" into the mount #2 and secure the lens mount to the telescope baseplate. Adjust the position of lens mount #2 so that the laser beam is centered on the target closest to the Infinity head. Adjust the translational position of the mount (with respect to the beam axis) so that the resulting beam diameter is as close as possible to the value given in Table 5-4.
7. Verify collimation of the laser beam by comparing burns taken at the laser head and at the target farthest from the laser system.

Verify that the collimation adjustment has not moved the laser beam off center on the target. If laser beam is found to be off center, return to Step 6.

8. Verify that the back reflection from lens #1 is striking the second dichroic mount on the stacked ceramics.

If the laser beam is centered, the installation and alignment of the Infinity telescope assembly is complete.

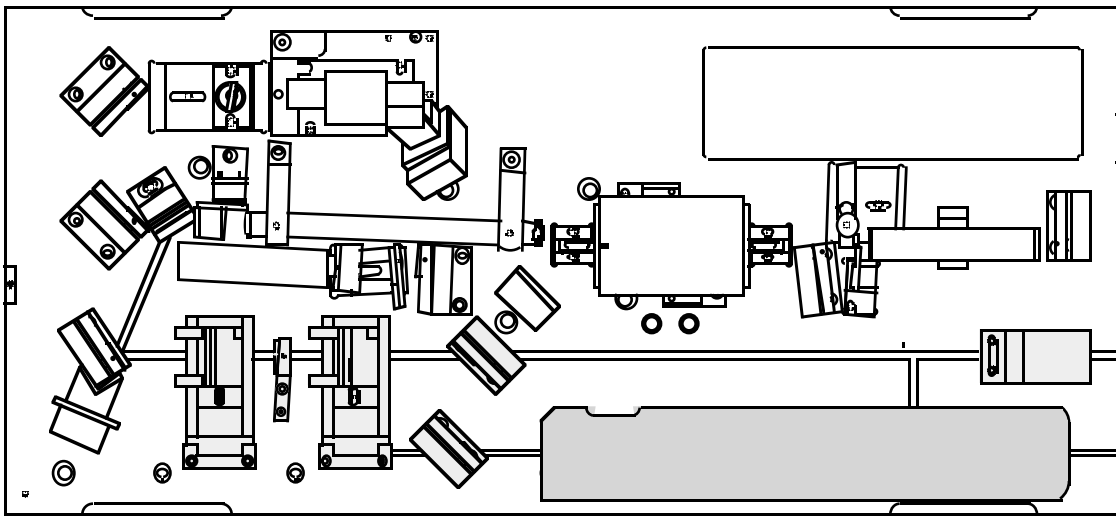


Figure 5-14. Orientation of steering dichroic mounts, IR beam dump, and Infinity Telescope base plate.

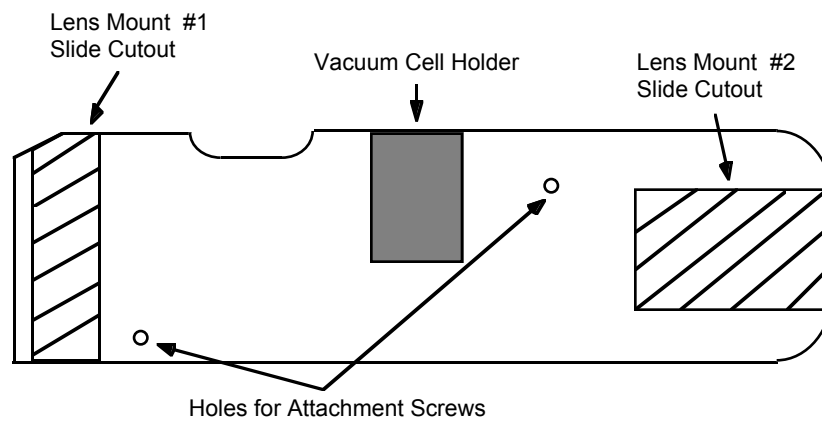


Figure 5-15. Overview of Telescope Mount Baseplate

